Analysis of Scientific Data – Semester 1, 2026

Applied Class 9 – Solutions

Part 1

Blood alcohol concentration depends on many factors including the number of standard drinks and body mass (kg). A study gave 21 participants various numbers of standard drinks and recorded their blood alcohol concentration (g/dL) from a urine sample after a fixed waiting period.

Any association between blood alcohol concentration and body mass may be obscured by the differences in the number drinks consumed. Assuming there is no interaction, the (edited) results from a multiple regression model for blood alcohol concentration with body mass and the number of drinks as explanatory variables are as follows:

```
lm(formula = BAC ~ Mass + Drinks, data = alcohol)

Coefficients:
             Estimate Std. Error
(Intercept)  0.0476753  0.0365179
Mass        -0.0001360  0.0004137
Drinks       0.0110798  0.0040792

Residual standard error: 0.01886 on 18 degrees of freedom
Multiple R-squared:  0.3276, Adjusted R-squared:  0.2529
```

a) Based on this model, what is the estimated blood alcohol concentration for a person with body mass 59 kg who has had 1.7 standard drinks?

Estimate is $\hat{y} = 0.0476753 - 0.0001360 \times 59 + 0.0110798 \times 1.7 = 0.0584$ g/dL.

b) Based on this model, is there any evidence of a negative association between BAC and body mass, after accounting for the number of standard drinks?

Test using

$$t_{18} = \frac{-0.001360 - 0}{0.0004137} = -0.329$$

One-sided p-value is $P(T_{18} \leq -0.329) = 0.373$, giving no evidence of a negative association between BAC and body mass, after accounting for the number of standard drinks.

c) Based on this model, what is the margin of error in a 95% confidence interval for the population parameter corresponding to the variable Drinks?

In general, margin of error is $t^* \times \text{se(estimate)}$. Here the estimate is the coefficient of Drinks in the model, with $\text{se}(b_2) = 0.0040792$ g/dL/drink.

The residual degrees of freedom are 18, so the critical value is $t^*_{18} = 2.100922$.

Together, the margin of error is $2.100922 \times 0.0040792 = 0.00857$ g/dL/drink.

Part 2

The following table shows counts of students in the 2024 and 2025 surveys by preferred superpower:

Year	Fly	Freeze Time	Invisibility	Telepathy	Total
2024	230	248	96	163	737
2025	211	165	93	129	598
Total	441	413	189	292	1335

Is there any evidence of a change in superpower preferences between 2024 and 2025?

Start by calculating totals, as shown. Expected value for "Fly" in 2024 is $\frac{737\times 441}{1335} = 243.5$, with similar calculations for remaining expected values (or by subtraction). This gives

$$\chi_3^2 = \frac{(230-243.5)^2}{243.5} + \frac{(248-228.0)^2}{228.0} + \cdots + \frac{(129-130.8)^2}{130.8} = 7.11$$

P-value is $P(\chi_3^2 \geq 7.11) = .0685$, weak evidence that the distribution of preferred superpower has changed between 2024 and 2025.

You may find the following output from R useful in answering the questions on this sheet.

```
> pt(3.48,df=8)
[1] 0.9958402
> pt(-0.329,df=18)
[1] 0.3729764
> qt(.95,df=18)
[1] 1.734064
> qt(.975,df=18)
[1] 2.100922
```

```
> pchisq(7.11,1)
[1] 0.9923345
> pchisq(7.11,2)
[1] 0.9714186
> pchisq(7.11,3)
[1] 0.9315269
> pchisq(7.11,7)
[1] 0.5824826
```